

VASUDEV BANSAL

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EDUCATION

2024 – 2028	B.Tech in Computer Science Engineering, USICT GGSIPU Delhi	CGPA: 8.1
2023	Class XII (CBSE), S.D. Public School	92%
2021	Class X (CBSE), S.D. Public School	91%

PROFILE

Computer Science undergraduate passionate about Backend Engineering, Distributed Systems and AI-powered applications. Experienced in building scalable full-stack applications, REST APIs, microservices and AI-integrated platforms using Node.js, React, Next.js, FastAPI and PostgreSQL through industry internships and production projects.

EXPERIENCE

Software Development Engineer (Full Stack) Intern — ChalksnBoard June 2026 – Present  chalksnboard.in

- Spearheaded the modernization of the **ChalksnBoard** website by migrating it to a scalable **React.js** and **Next.js** architecture, improving maintainability and brand presence.
- Enhanced user experience by implementing interactive scroll animations, responsive layouts and immersive web experiences using **Framer Motion**, **GSAP**, **Three.js** and **Tailwind CSS**.
- Designed reusable React components and modular UI sections, accelerating the development of landing pages and new website features.
- Improved website performance and technical SEO using **Next.js** server-side rendering, asset optimization and responsive design best practices.
- Developing **Vasooli AI**, an AI-powered fee collection platform using **Node.js**, **Express.js**, **MongoDB** and secure **JWT Authentication**.
- Built scalable **REST APIs** and integrated **WhatsApp (Interakt)**, **Twilio SMS**, **Redis**, **BullMQ** and Cron Jobs for automated communication workflows.
- Containerized backend services using **Docker** and explored multilingual AI voice technologies including **Microsoft Edge TTS**, **Coqui XTTS**, **ElevenLabs** and **LiveKit**.

Full Stack / AI Developer Intern — AICTE

June 2026 – July 2026

- Completed a 6-week internship developing **AgriYield**, an AI-powered smart agriculture platform using **React Native**, **Node.js**, **FastAPI** and **MongoDB**.
- Engineered a microservices architecture with **Node.js/Express** REST APIs and **Python FastAPI** services for real-time machine learning inference.
- Integrated **OpenWeather API** for GPS-based weather retrieval and **OGD India API** for live commodity market prices.
- Automated Android APK builds using **GitHub Actions**, **Expo CLI** and deployed backend microservices on **Render**.
- Collaborated in an agile team, performing code reviews, API testing with **Postman**, and unit/integration testing.

PROJECTS

Vasooli AI – AI Powered Fee Collection Platform

- Built an AI-powered fee collection platform using Node.js, Express.js, MongoDB and React Native.
- Developed secure JWT Authentication, REST APIs and role-based authorization.
- Integrated WhatsApp APIs, Twilio SMS, Redis and BullMQ.
- Used Docker for deployment and explored Microsoft Edge TTS, Coqui XTTS, ElevenLabs and LiveKit.

AgriYield – AI-Powered Smart Farming Platform

 [GitHub](#)

- Developed an end-to-end AI-powered smart farming platform using **React Native**, **Expo**, **Node.js**, **Express.js**, **Python FastAPI**, **MongoDB**, **Docker**, and **Render**.
- Built an **XGBoost** machine learning microservice exposed through **FastAPI**, analyzing 14 environmental and soil parameters to accurately predict crop yield.
- Implemented secure backend services with **JWT Authentication**, **bcrypt** password hashing, REST APIs, and MongoDB-based historical prediction storage.
- Integrated **OpenWeather API** for GPS-based weather data and **OGD India API** for real-time agricultural commodity prices to support data-driven farming decisions.
- Designed a responsive bilingual (**English/Hindi**) mobile application using **React Native**, **Expo**, and **react-i18next**, featuring a modern glassmorphism-inspired UI.
- Automated Android APK generation using **GitHub Actions** and **Expo CLI**, and deployed backend microservices on **Render**.

AssetFlow – Enterprise Asset & Resource Management System

 [Live Demo](#)

- Designed and deployed a full-stack enterprise asset management platform using **Next.js**, **PostgreSQL**, **Prisma ORM** and **Supabase** to centralize asset tracking, maintenance and resource allocation.

- Implemented secure **JWT Authentication** with strict **Role-Based Access Control (RBAC)** supporting four user roles: Admin, Manager, Department Head and Employee.
- Engineered a conflict-free shared resource booking system with approval workflows, transfer requests and calendar scheduling to eliminate double-booking.
- Built scalable **REST APIs**, real-time KPI dashboards and optimized database operations using **Next.js Route Handlers** and **Prisma ORM**.
- Developed a responsive enterprise-grade UI using **CSS Modules**, Glassmorphism design principles and an Odoo-inspired design system.

TECHNICAL SKILLS

Languages	C++, Java, Python, JavaScript, SQL, HTML/CSS
Frontend Development	React.js, Next.js, TypeScript, Tailwind CSS, HTML5, CSS3, JavaScript, React Native (Expo), Framer Motion, GSAP, Three.js
Backend Development	Node.js, Express.js, Python FastAPI, REST APIs, JWT Authentication, Middleware, API Integration, Prisma ORM, BullMQ, Cron Jobs
Databases	PostgreSQL, MongoDB Atlas, MySQL, SQLite, Redis, Supabase
AI Technologies	Gemini API, XGBoost, FastAPI, Microsoft Edge TTS, Coqui XTTS, ElevenLabs, LiveKit
DevOps & Tools	Docker, Git, GitHub Actions, Postman, Render, Vercel, Redis, BullMQ, Expo CLI

ACHIEVEMENTS

- Solved **250+** LeetCode problems across Arrays, Trees, Graphs, Dynamic Programming and Binary Search.
- Built multiple production-ready full-stack and backend applications using React, Next.js, Node.js, MongoDB, PostgreSQL and RESTful APIs.
- Hands-on experience with Docker, Redis, BullMQ, AI voice technologies and scalable backend architecture.
- Strong interest in Backend Engineering, Distributed Systems and AI-powered applications.